

VIETNAM NATIONAL UNIVERSITY OF HOCHIMINH CITY  
THE INTERNATIONAL UNIVERSITY  
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



# **An LLM-Augmented Approach for Topic Modeling and Emotion Recognition in Vietnamese Online Education Platforms**

By  
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A thesis submitted to the School of Computer Science and Engineering  
in partial fulfillment of the requirements for the degree of  
Bachelor of Information Technology/Computer Science/Computer Engineering

Ho Chi Minh City, Vietnam  
2025

# **An LLM-Augmented Approach for Topic Modeling and Emotion Recognition in Vietnamese Online Education Platforms**

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## **ABBREVIATIONS**

IT: Information Technology

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## ABSTRACT

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# CHAPTER 1

## INTRODUCTION

### 1.1. Background

The past few years have witnessed a drastic advancement in technology, enhancing nearly every aspect of life. Education, in particular, is a field that has seen a significant shift toward online learning. For this reason, the need for a system that can understand student feedback and concerns has become increasingly important for the education system. Monitoring discussion forums and analyzing student's feedback and comments is needed for gaining valuable insights, from which can be used to tailor the teaching and management of the education system.

#### 1.1.1. Challenges in Vietnamese Natural Language Processing (NLP)

Vietnamese is considered a low-resource language in natural language processing (NLP) research, making the implementation of language analysis systems such as sentiment analysis and topic modeling quite a difficult task. Historically, research on Vietnamese sentiment analysis (SA) has been hampered by the limited availability of suitable datasets. Existing educational datasets for Vietnamese NLP often lack sufficient domain relevance and student slang, which presents a significant challenge for conducting research and improving the educational experience.

#### 1.1.2. The demand for Advanced Text Analysis in Education

The two core methods of Text Analysis are Topic Modeling and Sentiment Analysis. These are essential for extracting meaning from the enormous volume of student-generated content, such as feedback, posts, and comments on online platforms.

## **Topic Modeling and Classification:** Text classification

### **1.2. Problem Statement**

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### **1.3. Scope and Objectives**

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### **1.4. Assumption and Solution**

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### **1.5. Structure of thesis**

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## **CHAPTER 2**

### **LITERATURE REVIEW/RELATED WORK**

#### **2.1. Review 1**

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## **CHAPTER 3**

### **IMPLEMENTATION**

#### **3.1. User requirement analysis**

Abc...

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Abc...

#### **3.2. System Design**

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##### **3.2.1. Database design**

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#### **3.3. Abc**

## CHAPTER 4

### EXPERIMENTS AND RESULTS

#### 4.1. Experiments

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##### 4.1.1. Abc

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Abc...

#### 4.2. Results

Abc...

##### 4.2.1. Result 1

Abc...

##### 4.2.2. Result 2

Abc...

## **CHAPTER 5**

### **DISCUSSION AND EVALUATION**

#### **5.1. Discussion**

Abc...

#### **5.2. Comparison**

Abc...

#### **5.3. Evaluation**

Abc...



## **CHAPTER 6**

### **CONCLUSION AND FUTURE WORK**

#### **6.1. Conclusion**

Abc...

#### **6.2. Future work**

Abc...

## REFERENCES

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## APPENDICES